

CodeNova Business Website

Database Design Document

Version: 1.0 **Project:** CodeNova Business Website

Defines the recommended PostgreSQL/Supabase data model, relationships, indexes, security policies and data lifecycle for the CodeNova platform.

1. Database Platform

- PostgreSQL through Supabase is recommended.
- Use migrations for all schema changes.
- Use UTC timestamps in storage where practical.
- Use UUID primary keys for externally referenced records.

2. users

- id UUID PK.
- name VARCHAR.
- email VARCHAR UNIQUE.
- role VARCHAR/enum.
- auth_user_id UUID/reference where applicable.
- created_at TIMESTAMP.
- updated_at TIMESTAMP.

3. leads

- id UUID PK.
- name VARCHAR NOT NULL.
- email VARCHAR NOT NULL.
- phone VARCHAR.
- company VARCHAR.
- country VARCHAR.
- service VARCHAR.
- budget VARCHAR.
- message TEXT.
- source VARCHAR.
- status VARCHAR/enum.
- notes TEXT.
- assigned_to UUID nullable.
- created_at TIMESTAMP.
- updated_at TIMESTAMP.

4. consultations

- id UUID PK.
- lead_id UUID FK → leads.
- preferred_date DATE.
- preferred_time TIME.
- status VARCHAR.
- meeting_url TEXT.
- notes TEXT.
- created_at TIMESTAMP.
- updated_at TIMESTAMP.

5. estimator_requests

- id UUID PK.
- lead_id UUID nullable FK.
- project_type VARCHAR.
- complexity VARCHAR.
- selected_features JSONB.
- estimated_min NUMERIC.
- estimated_max NUMERIC.
- created_at TIMESTAMP.

6. services

- id UUID PK.
- slug VARCHAR UNIQUE.
- name VARCHAR.
- description TEXT.
- icon VARCHAR.
- status BOOLEAN.
- sort_order INTEGER.
- created_at TIMESTAMP.
- updated_at TIMESTAMP.

7. portfolio_projects

- id UUID PK.
- slug VARCHAR UNIQUE.
- name VARCHAR.
- category VARCHAR.
- description TEXT.
- technologies JSONB.

- cover_image TEXT.
- content JSONB/TEXT.
- status BOOLEAN.
- created_at TIMESTAMP.
- updated_at TIMESTAMP.

8. blog_posts

- id UUID PK.
- slug VARCHAR UNIQUE.
- title VARCHAR.
- excerpt TEXT.
- content TEXT.
- cover_image TEXT.
- category VARCHAR.
- author UUID/reference.
- seo_title VARCHAR.
- seo_description TEXT.
- status VARCHAR.
- published_at TIMESTAMP nullable.
- created_at TIMESTAMP.
- updated_at TIMESTAMP.

9. testimonials & settings

- testimonials: id, client_name, company, role, content, rating, image, status, created_at.
- site_settings: id, key UNIQUE, value JSONB/TEXT, updated_at.

10. Relationships

- leads 1→many consultations.
- leads 1→many estimator_requests.
- users/admins 1→many assigned leads.
- Services, portfolio and blog use unique slugs for public routes.

11. Indexes

- Index leads(status).
- Index leads(created_at).
- Index leads(service).
- Index leads(email) where useful.
- Unique indexes on services(slug), portfolio_projects(slug), blog_posts(slug).
- Add indexes based on actual query performance rather than indexing every column.

12. RLS / Security

- Enable Row Level Security for exposed Supabase tables where appropriate.
- Public users should not be allowed to read the entire leads table.
- Public insert policies should be narrowly scoped to required form operations or, preferably, performed through secure server endpoints.
- Admin read/write access must be role-restricted.
- Never expose service-role credentials in the browser.

13. Data Retention

- Define retention periods for leads, inquiries, analytics-related records and admin logs.
- Delete/anonymize records when no longer needed, subject to legal and business requirements.
- Back up production data and periodically test restoration.

Document Relationship

This document complements the CodeNova PRD, TRD, Security & Privacy Document and Frontend Requirements Document. Implementation should use the approved business information and actual production configuration rather than placeholders or unverified claims.